

Ways to declare 2D arrays

psvm(string... args)

```
{
    int [][] arr; // preferred
    int [] arr[];
    int Dim [][];

    int [] ab, cd; // ab, cd 1D array
    int [] x[], y; // x 2D and y 1D array
    int [][] x[], y; // x 3D and y 2D array
    int [] [] x, y; // x 2D and y 2D array
    int [] [] aa, bb[]; // aa 2D and bb 3D
    int [] aa, [] bb (X) CE
    int [] aa, bb[]; (V)
```

- when to use array
- when u want to store ^{similar} ~~single~~ type of data with fixed size and fast access and modification of data

• Arrays are treated as objects.

array $\begin{cases} \nearrow \text{Primitive data} \\ \searrow \text{Object data} \end{cases}$

collection $\rightarrow$ object data

```
int [] arr = new int [Size]
```

byte, $\sqrt{\text{int, short, char}}$

Exception:

- (i)

```
int [] arr = new int [2];
arr [2]; // Exception index out of bound
```
- (ii)

```
int [] arr = new int [-2]; // Exception
-ve array size exception
```
- (iii)

```
int [] arr = new int [2];
arr [0] = "sch"; // Exception array store exception
```

i) `int[] a = new int [9999 - 9]`

Exception array out of memory

- Every type of array will have specific class (inbuilt class) not available for programmer
for java lang, not for programmer for programmer

- static methods are there in Arrays class

ex-

- `Arrays.sort(arr name)`
- `Arrays.fill(arr, data)` etc

transfer statements

=====

1. break
2. continue
3. return

Q>

```
class Test{
    public static void main(String args[]){
        int x=0;
        switch(x){
            case 0: System.out.println("hello");
                    break ;//It is used to avoid fallthrough in switch
            case 1: System.out.println("hi");
        }
    }
}
```

Predict the Output:

- A. Compile Time Error
- B. Some problem occurred by jvm during execution
- C. hello
hi
- D. hello

Answer: D

Q>

```
class Test{
    public static void main(String args[]){
        for(int i=0; i<10; i++) { // i = 0, 0<10(true), i =
1, 1<10(true)
                                if(i==5)
                                break; //control will come out of the executing
loop
                                System.out.print(i); //0,1,2,3,4
        }
    }
}
```

Predict the Output:

- A. Compile Time Error
- B. Some problem occurred by jvm during execution
- C. 0 1 2 3 4
- D. 0 1 2 3 4 6 7 8 9

Answer: C

Q>

```
class Test{
    public static void main(String args[]){
        int x=10;
        l1 : {
                                System.out.println("begin");
                                if(x==10)
                                break l1;
                                System.out.println("end");
        }
        System.out.println("hello");
    }
}
```

```

        }
    }
}

```

Predict the Output:

- A. Compile Time Error
- B. Some problem occurred by jvm during execution
- C. begin
end
- D. begin
end
hello
- E. None of the above

Answer: E(begin and hello will be printed)

```

Q>
class Test{
    public static void main(String args[]){
        int x=10;
        if(x==10)
            break;
        System.out.println("hello");
    }
}

```

Predict the Output:

- A. Compile Time Error
- B. Some problem occurred by jvm during execution
- C. hello
- D. None of the above

Answer: A (break can be used in switch, loop and labelled block, other place compile time error)

```

Q>
class Test{
    public static void main(String args[]){
        int x=2;
        for(int i = 0; i<10;i++) {
            if(i%x==0)
                continue;//it will skip the current iteration and proceeds with
next iteration
            System.out.println(i);
        }
    }
}

```

Predict the Output:

- A. Compile Time Error
- B. Some problem occurred by jvm during execution
- C. 0 2 4 6 8
- D. 1 3 5 7 9
- E. None of the above

Answer: D

```

Q>
class Test
{
    public static void main(String args[]){

```

```

        int x=10;
        if(x==10)
        continue;//continue can't be used here
        System.out.println("hello");
    }
}

```

Predict the Output:

- A. Compile Time Error
- B. Some problem occurred by jvm during execution
- C. hello
- E. None of the above

Answer: A

Q>

```

class Test{
    public static void main(String args[]){
        int x=0;
        switch(x){
            case 0: System.out.println("hello");
                    continue;
            case 1: System.out.println("hi");
        }
    }
}

```

Predict the Output:

- A. Compile Time Error
- B. Some problem occurred by jvm during execution
- C. hello
hi
- D. hello

Answer: A(continue can be used only in loops and labelled block, other places compile time error)

Note:

```

l1:
for()
{
    l2: for()
    {
        l3: for()
        {
            break/break l3; // goto stmt-1
            break l2; //goto stmt2
            break l1; // goto stmt3
        }
        stmt-1;
    }
    stmt-2;
}
stmt-3;

```

Q>

```

class Test{
    public static void main(String args[]){

```

```

        l1:for(int i=0;i<3;i++) // i = 0 ,0<3(true) , i =1, 1<3(true), i
=2 , 2<3(true), i = 3 , 3<3(false)
        for(int j=0;j<3;j++)// j =0,0<1(true), j =1,
1<3(true),j =2 ,2<3(true)
        {
            if(i==j)
                stmt1;

            System.out.println(i+""+j);//10 20 21
        }
    }
}

```

replace stmt1 with break and predict the output?

Answer:1 0 2 0 2 1

replace stmt1 with break l1 and predict the output?

Answer: no output

replace stmt1 with continue and predict the output?

Answer: 01, 02,10,12,20,21

replace stmt1 with continue l1 and predict the output?

Answer: 10 20 21

Q>

```

class Test{
    public static void main(String args[]){
        while(true){
            System.out.println("hello");//line-n1
        }
        System.out.println("hi");//line-n2
    }
}

```

Predict the output

- A. Compile time error at line-n1
- B. some problem occurred during jvm execution.
- C. Compile time error at line-n2
- D. hello
hi
- E. hello infinite times
- F. None of the above

Answer: C

Q>

```

class Test{
    public static void main(String args[]){
        if(true){
            System.out.println("hello");//line-n1
        }
        else{
            System.out.println("hi");//line-n2
        }
    }
}

```

Predict the output

- A. compile time error at line-n1

- B. some problem occurred during jvm execution.
- C. Compile time error at line-n2
- D. hello
 hi
- E. hello
- F. hi
- G. None of the above

Answer: E

Concept of unreachable holds good only for loops(for,while,dowhile), compiler will ignore unreachable for if else and switch syntax.

```
Q>
class A
{
    void add(byte a, byte b)
    {
        //stmt;
    }
}

class Test
{
    public static void main(String[] arg)
    {
        A obj=new A();
        obj.add(10,20); //Line 1 => CE: Why it is throwing CE. The value 10 is
inbetween byte range(-128 to 127 )

        int i1=10;
        byte b1=10; // Line 2 => Here Compiler accepting the value 10.
        byte b2=i1; //Line 3 => CE
    }
}
```

